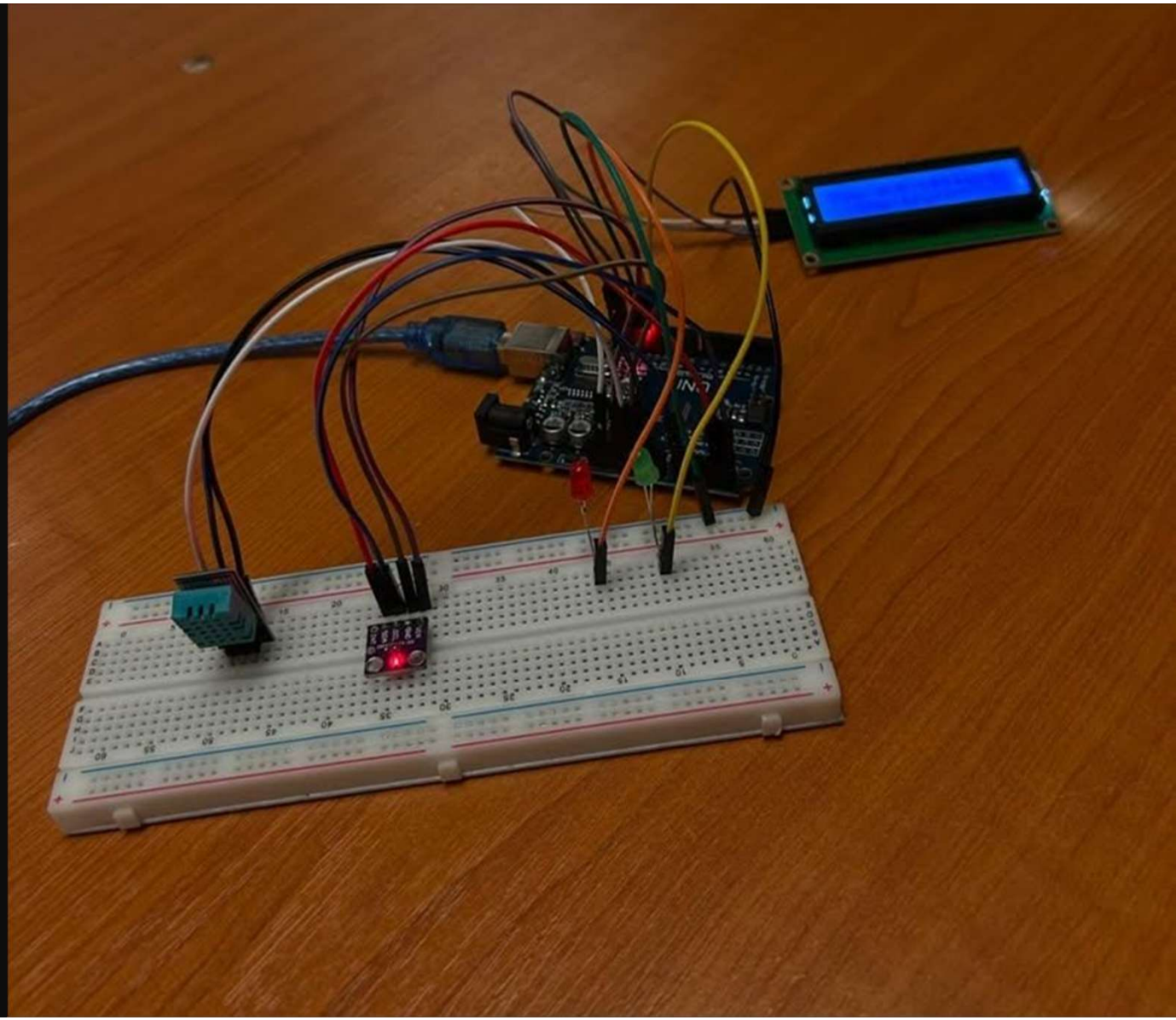
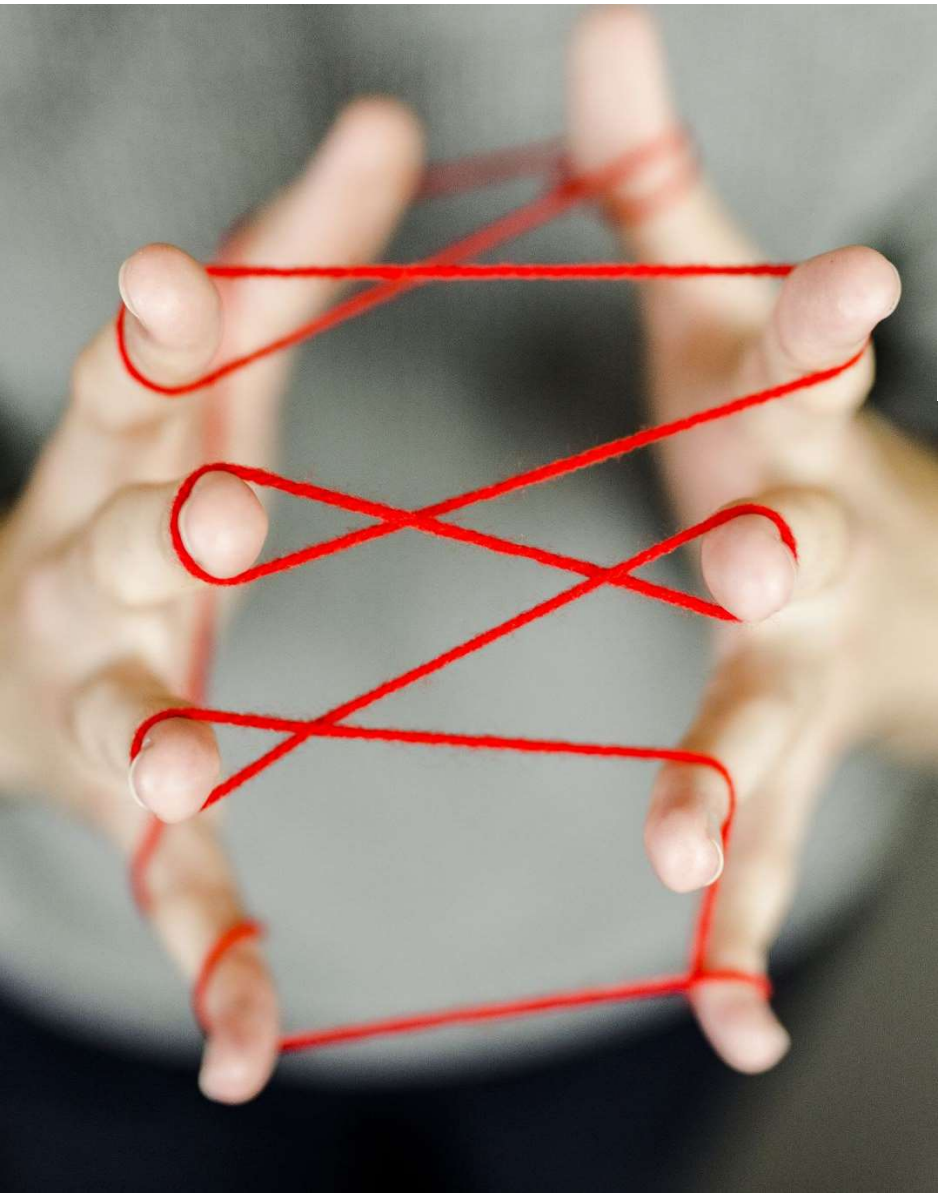
Lie Detector

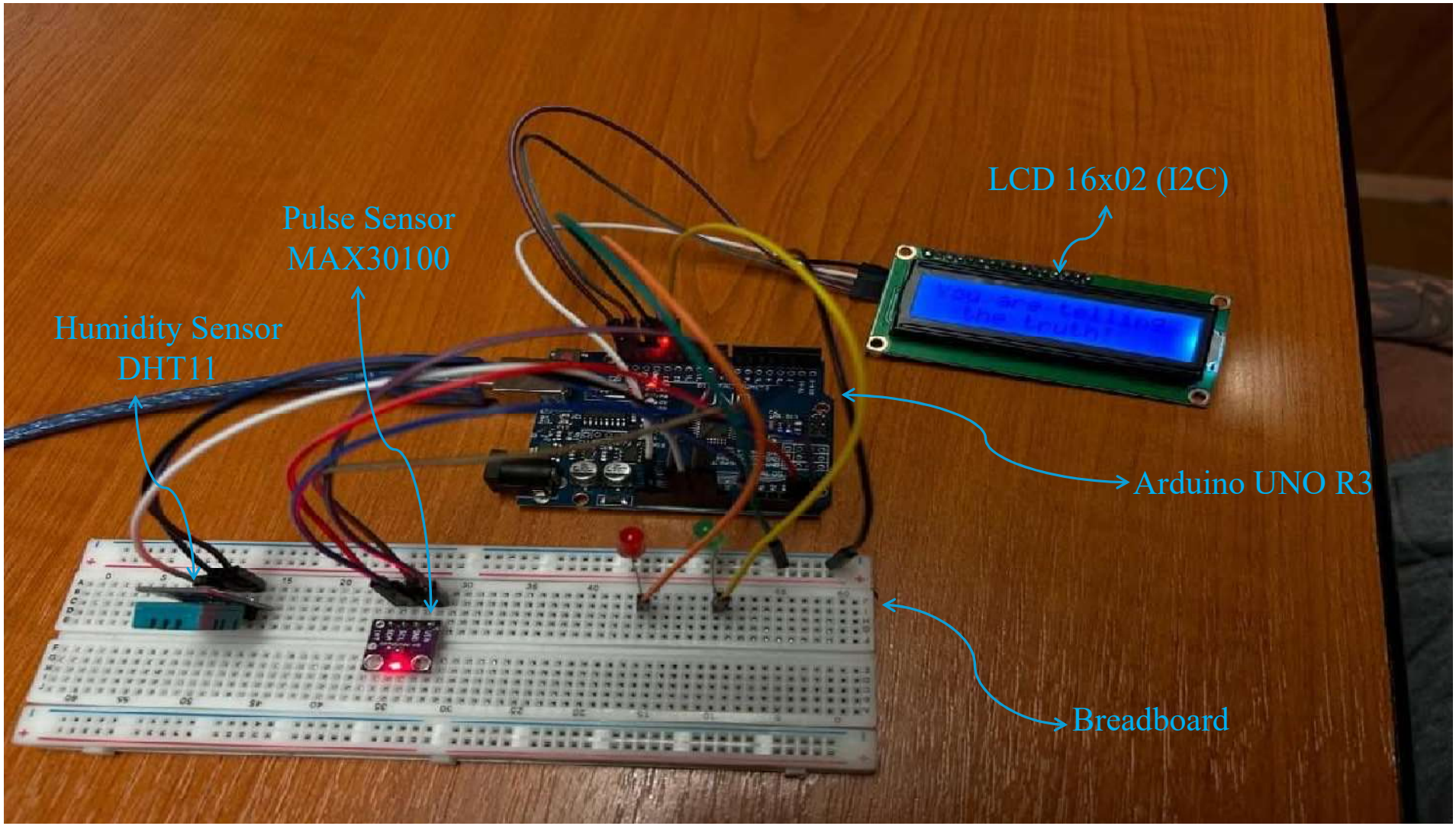
BY: Nichita Marcus and Ruba Astrid





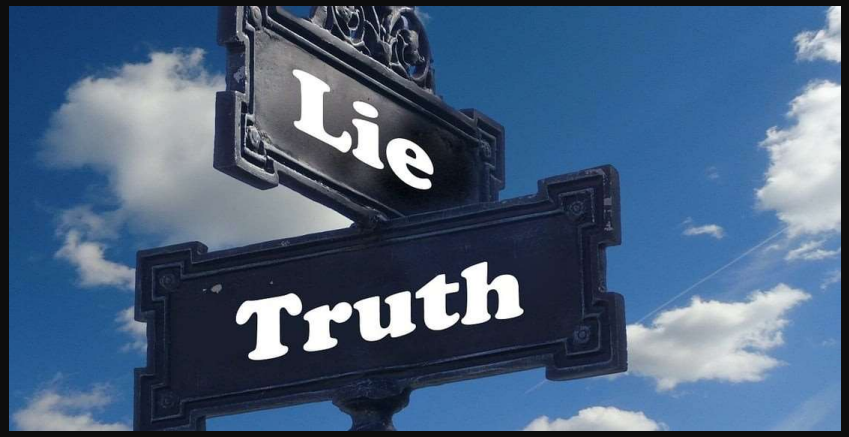
Introduction

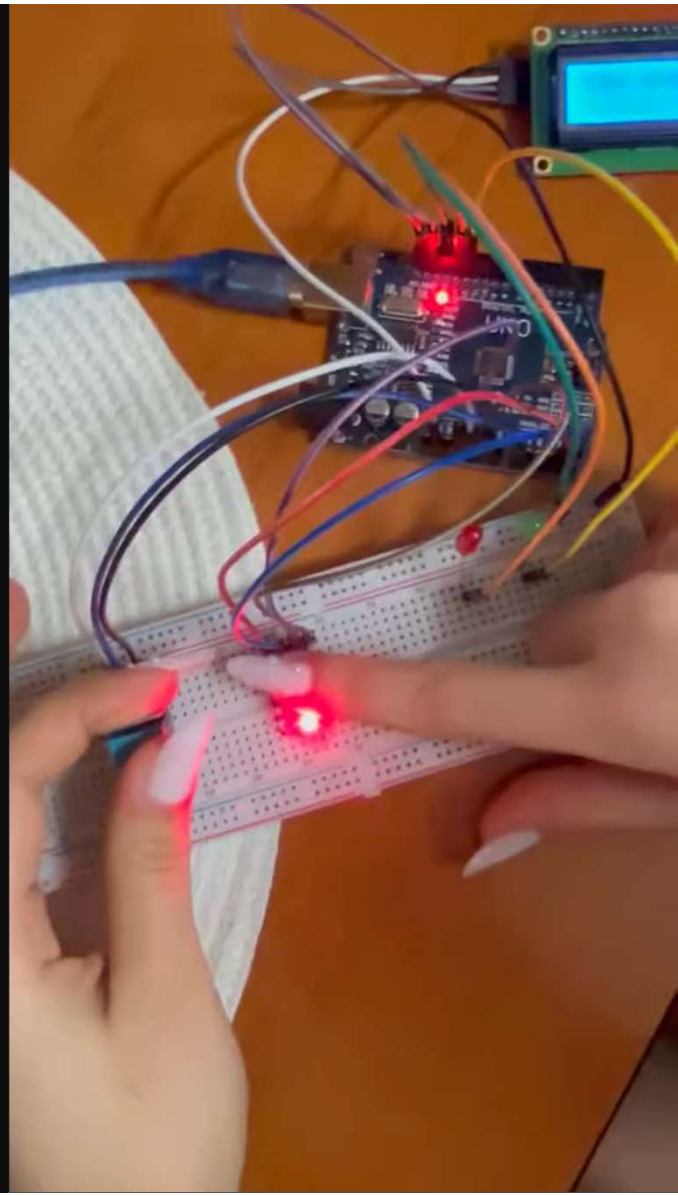
- Our project represents a lie detector based on the humidity of your fingers and on the BPM(Beats Per Minute) of your heart.
- We chose this idea because it seemed interesting, and we saw it as an opportunity for something cool.
- Although the sensors are a bit faulty, by fine tunning the threshold values the detector can be quite accurate.


LCD 16x02 (I2C)
Pulse Sensor
MAX30100
Humidity Sensor
DHT11
Arduino UNO R3
Breadboard

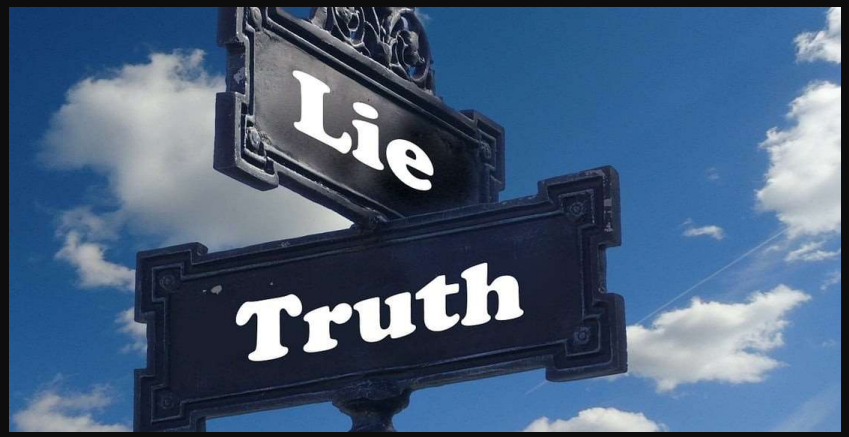
Video Showcase

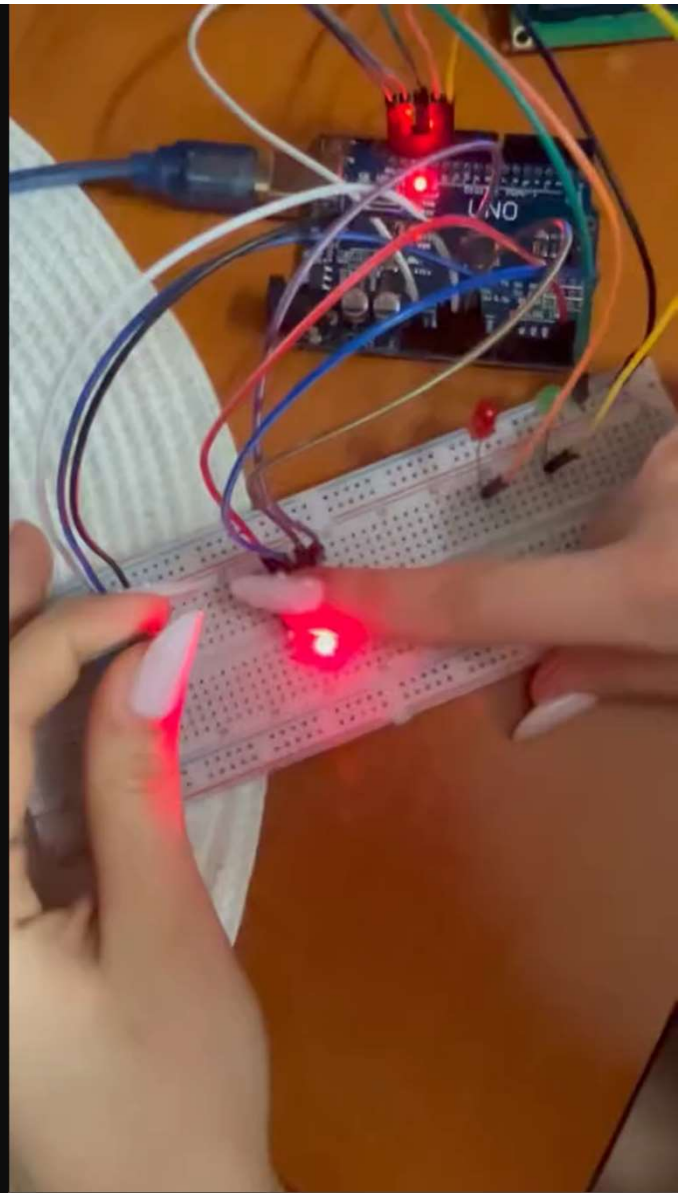
Truth



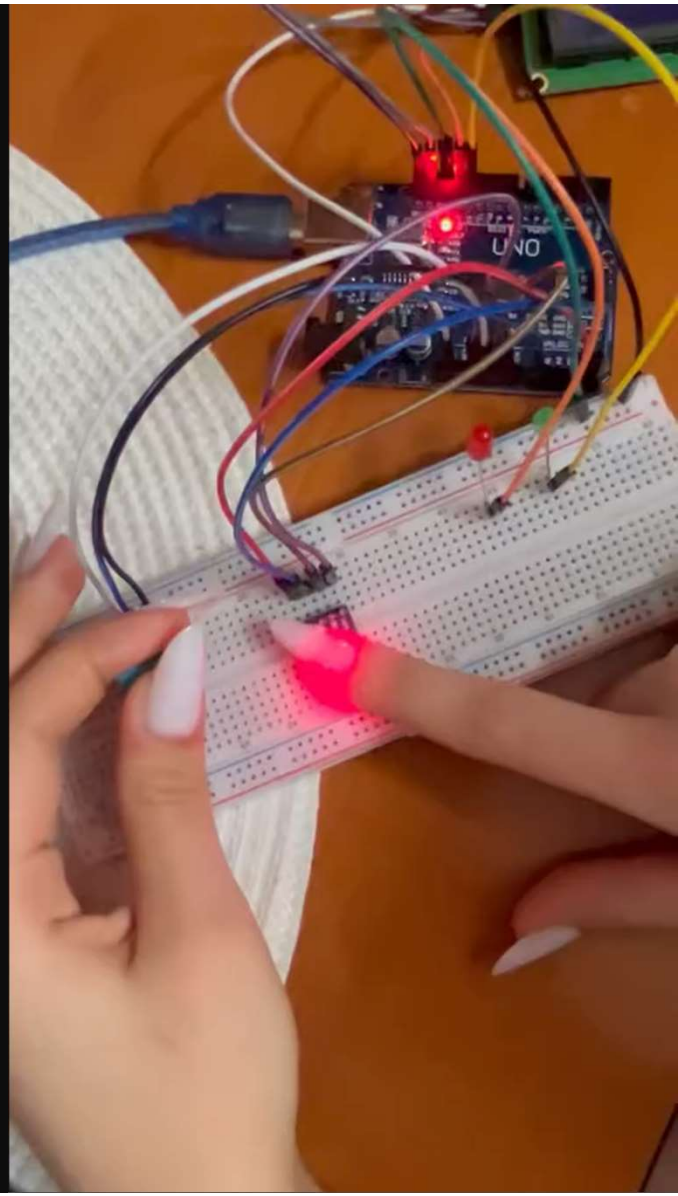


Lie





Improper Use



Code Showcase

```
// Continuă măsurătoarea dacă Heart_rate nu este 0
if (is_measurement_in_progress) {
    if (millis() - startTime <= 15000) {
        Humidity = dht.readHumidity();
        Temperature = dht.readTemperature();

        // Verifică validitatea datelor
        if (isnan(Humidity) || isnan(Temperature)) {
            Serial.println("Eroare la citirea senzorului DHT!");
            return;
        }

        // Afișează pe Serial Monitor
        Serial.print("Heart rate: ");
        Serial.println(Heart_rate);
        Serial.print("Humidity: ");
        Serial.print(Humidity);
        Serial.println(" %");
        Serial.print("Temperature: ");
        Serial.print(Temperature);
        Serial.println(" °C");

        // Afișează pe LCD
        lcd.setCursor(0, 0);
        lcd.print("HR: ");
        lcd.print(Heart_rate);
        lcd.print(" bpm");
```

Masurarea o data la 15 secunde si afisarea pe ecran

Code Showcase

```
// Funcție pentru a începe măsurarea de 10 secunde
void startMeasurement()
{
    pulse_sum = 0;
    humidity_sum = 0;
    count = 0;
    is_measurement_in_progress = true;
    startTime = millis();
}

// Resetează variabilele pentru măsurare
void resetMeasurement()
{
    is_measurement_in_progress = false;
    pulse_sum = 0;
    humidity_sum = 0;
    count = 0;
}
```

Functiile pentruu inceperea masurarii si resetarea masuratorilor

Thank you for your attention